

SIGNIFICANCE

The unmet need. If we are fortunate, each of us will reach the point where we, or someone we love, needs help with the ordinary tasks of daily living: bathing, dressing, meals, medications, and moving safely through the home. Most families meet that moment at its worst, after a fall, a diagnosis, or a sudden hospitalization, and are expected to become experts in eldercare overnight.

A recognized need becomes established care only if a family clears three gates (Figure 1). They must **find** the right care inside a fragmented ecosystem of public aid programs, insurance coverage, healthcare services, and long-term services and supports, each with separate eligibility rules and none accountable for whether care begins. They must **afford** it: full-time home care runs \$80,080 a year,² Medicare does not cover custodial home care, and Medicaid, the nation's largest payer of long-term care, pays only after a family has spent down to eligibility. An estimated \$58 billion in assistance that would close part of that gap goes unclaimed every year because families cannot identify or navigate the programs.¹ And someone must be available to **deliver** the care: 63.3 percent of home-care providers declined cases in 2023 that they could not staff.³

Failure at any one gate produces the same outcome. Needs go unmet, function declines, a crisis arrives with an emergency visit or a hospitalization, and discharge returns the older adult home without the support that was missing in the first place. Nearly a third of older adults with difficulty in activities of daily living report going without bathing, meals, or medications in a given month,⁴ and unmet needs of this kind independently predict emergency department use, readmission, nursing home placement, and mortality.^{5,6,7} Families of modest means therefore wait, and decline care, until a hospitalization, when public programs finally step in at the most expensive possible moment and after the most harm has already occurred.

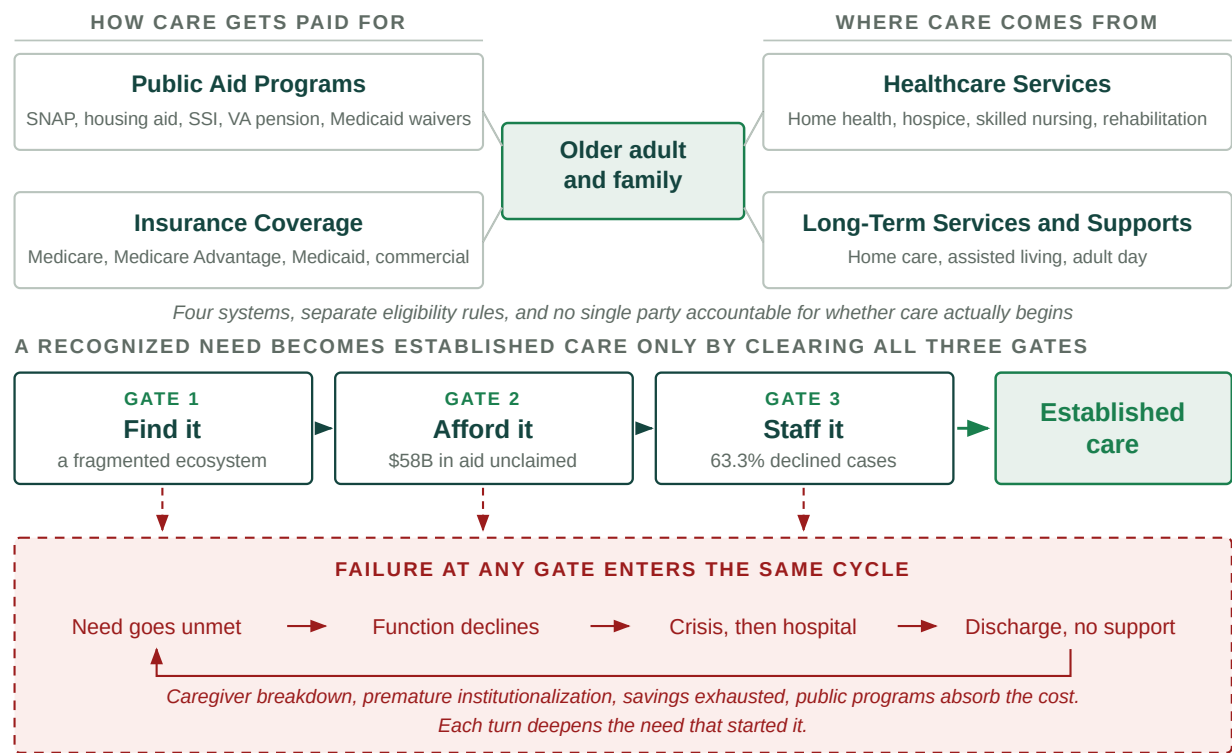


Figure 1. The eldercare ecosystem a family must navigate, the three gates that must all be cleared for a recognized need to become established care, and the cycle that follows failure at any one of them.

What it costs, and who pays it. The gap is widening from both sides. The population over 65 reaches 82 million by 2050,⁸ while the family capacity that has always absorbed most of American eldercare is contracting: the number of potential family caregivers for each adult over 80 falls from more than seven in 2010 to four by 2030 (Figure 2).⁹ The paid workforce that would have to replace it faces 9.7 million direct-care job openings between 2024 and 2034.¹⁰ Demand is rising against two supplies that are falling at once.

The costs of that gap do not disappear. They move, and they grow in transit. Roughly 70 percent of working-age family caregivers hold jobs, and the hours they lose are borne by employers and by their own earnings.¹¹ Families spend down their savings and Medicaid inherits the bill at the institutional rate. The clearest

measurement of the inversion comes from CAPABLE, a home-based, function-focused support program whose implementation was associated with roughly \$3,000 in program cost against more than \$20,000 in reduced Medicaid spending per participant, driven by lower inpatient and outpatient use.¹² The same approach reduced disability across multiple randomized trials.¹³

The money to support older adults at home largely exists already. It is simply spent later, on worse outcomes, by whoever is left holding the consequence. Every dollar of that consequence sits on an identifiable balance sheet: a health plan's, a state Medicaid budget's, a health system's, an employer's, or a family's. That is what makes this an addressable market and not only a public health problem.

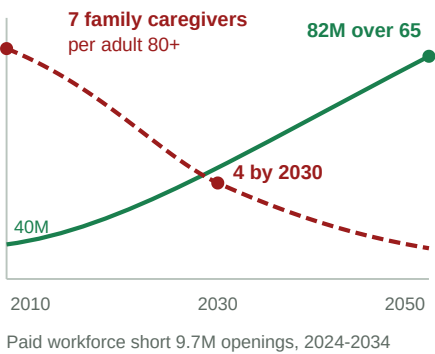


Figure 2. Demand rising while both sources of supply contract.

What establishing care actually requires. Closing the gap is not a matter of better information. A family that knows exactly which program it qualifies for still has to complete the application, gather documents, file, wait, respond to requests, and then find a provider with capacity on the date the benefit starts. Existing help stops before that point: clinicians and discharge planners hand off, information services return lists, and referral marketplaces introduce a provider and are paid on placement. None is accountable for whether care was established, and none can create a caregiver who does not exist.

Why navigation alone cannot close the gap. Two things are required at once: navigation that runs all the way to a confirmed start, and capacity on the other side so that the start is possible. A product that clears only the first two gates aggregates family demand and delivers it to a supply side that cannot absorb it, which is a faster path to the same refusal. That is not hypothetical; it is why nearly two-thirds of home-care providers turned away business in 2023.³ Olera did not select workforce development as an adjacent opportunity. We encountered it as the binding constraint on our own product working, which is why capacity creation sits inside the product rather than beside it.

The product and the path to market. CareNavigator is an eldercare AI navigation system built over an expert-curated national database of aid programs and providers, developed across NIA SBIR Phases I through IIB. It screens a household's needs and means and identifies the aid and services that fit. What this award adds is execution: agents that file the application or send the intake packet, follow up on a schedule, and confirm the date aid or care began. Securing a benefit and securing a service differ in their steps, and the system adapts to which is being pursued, but the shape is the same: prepare the request, transmit it, follow up, confirm the outcome. Alongside it, Olera recruits and verifies new caregivers and places them with licensed providers, who hire, train, insure, and supervise them as employees.

Olera builds two products, CareNavigator and Caregiver Staffing. During this award only Caregiver Staffing is sold and priced. Families use CareNavigator at no cost, and no provider pays to be listed or to receive a family connection. Because no fee is charged for a referral, no steering incentive exists and federally reimbursed providers can participate, which pay-per-referral models cannot accommodate.

Market segments and customers. The beachhead customer is the healthcare and long-term services provider. Roughly 165,000 such organizations operate in the United States,^{15,16} and they already spend at least \$5 billion a year on marketing and caregiver recruiting, an estimate built entirely from published low-end inputs.¹⁷ They buy from us for an auditable reason. The published median cost of acquiring a caregiver is \$520 per hire, and that figure is for word-of-mouth, the cheapest channel an agency has.¹⁸ Table 1 sets our pricing against that benchmark on the provider's own numbers.

Agency size	Hires per year at 75% turnover	Current cost at the \$520 median	Olera at \$275/mo, unlimited hiring	Effective cost per hire	Annual saving
15 caregivers	11	\$5,850	\$3,300	\$293	\$2,550
30 caregivers	23	\$11,700	\$3,300	\$147	\$8,400
50 caregivers	38	\$19,500	\$3,300	\$88	\$16,200

Agency size	Hires per year at 75% turnover	Current cost at the \$520 median	Olera at \$275/mo, unlimited hiring	Effective cost per hire	Annual saving
100 caregivers	75	\$39,000	\$3,300	\$44	\$35,700

Table 1. What a provider pays today to hire a caregiver, and what Olera charges instead.

We do not ask providers to believe in a new category. We ask them to pay less for something they already buy. Break-even against the benchmark arrives at 6.3 hires a year, which is an agency of roughly eight caregivers, so every larger agency is ahead from the first month. The revenue side is larger still: at 63.3 percent of agencies declining cases they cannot staff, the opportunity cost of an unfilled role dwarfs the subscription price.

A second customer class is emerging and this award evidences rather than sells to it. Medicare now pays directly for care navigation, caregiver support, and respite through the GUIDE model, which began in July 2024.¹⁹ Medicare Advantage plans, Medicaid managed care organizations, health systems and accountable care organizations, self-insured employers, and long-term care insurers each bear part of the cost of unmet need. Each bears part of the cost of unmet need, and the establishment and cost data this award produces is what makes them addressable. **None of them is tested in this award and none carries an endpoint in any aim.** They are modeled in the Commercialization Plan, where the nearest-term line is the provider growth tools Olera already operates.

Competitive environment and our advantage. Table 2 places CareNavigator against every category of alternative across the five stages of the path to care. The pattern rather than any single row is the argument: **every alternative solves one stretch of the path and stops at a handoff, and the handoffs are where families are lost.** Human navigators are the strongest competitor and cover four of the five stages; they are also scarce, episodic, and cannot be scaled by hiring, which is the same workforce constraint from a different direction. Staffing channels move workers between employers, so one provider's hire is another's vacancy and the national shortage does not shrink. Referral marketplaces charge placement fees, which determines who appears and has drawn sustained criticism of the incentives it creates.²²

Stage of the path to care	CareNavigator navigation and new workforce	Public information <i>Eldercare Locator, BenefitsCheckUp</i>	General AI assistants <i>ChatGPT, Claude, Gemini</i>	Human navigators <i>social workers, discharge planners</i>	Referral marketplaces <i>A Place for Mom, Caring.com</i>
Assesses needs	✓	✓	✓	✓	✓
Identifies aid and insurance	✓	✓	✓	✓	✗
Matches local providers	✓	✗	✗	✓	✓
Carries the case to established care	✓	✗	✗	✓	✗
Adds the caregivers to staff it	✓	✗	✗	✗	✗

Table 2. Coverage of the path to care: the alternatives families use.

Caregiver Staffing competes in a separate market with its own incumbents, and Table 3 places it against them. Job boards leave every screening decision to the provider, staffing agencies keep the worker as their own employee, and gig platforms fill a shift without filling a role. All three draw from the same constrained pool of people already in direct care, which is the distinction the last row records and which Key Innovation 1 develops.

What a provider needs from a hiring channel	Olera Caregiver Staffing	Job boards <i>Indeed, MyCNAJobs</i>	Staffing agencies <i>local and franchise</i>	Gig platforms <i>shift marketplaces</i>
Delivers applicants at low cost	✓	✓	✗	✓
Screens and verifies before the provider spends time	✓	✗	✓	✗
Worker becomes the provider's own employee	✓	✓	✗	✗
Adds workers who were not already in direct care	✓	✗	✗	✗

Table 3. Coverage of the hiring problem: the channels providers use today.

Related development efforts in academia and industry. Most of the relevant science is settled and what remains open is operational, which is the reason this is a commercialization award rather than a trial. Research on unmet need, from the National Health and Aging Trends Study and Medicare-linked cohorts, establishes that unmet daily-activity needs predict downstream utilization and placement;^{4,5} we treat that as settled and measure care established rather than clinical endpoints. Home-based support trials, principally CAPABLE, show that function-focused support reduces disability with savings well above program cost,^{12,13} but no trial addresses how a family outside a trial finds such support, funds it, and gets it started, and CareNavigator is the front door those programs assume exists. Work on digital caregiver tools, including four peer-reviewed studies of our own, establishes that such platforms can be built and measured as usable and accepted;^{25,26,27,28} what remains open is whether agents can complete applications and carry a case to a decision. Payer-funded navigation is no longer speculative, given the GUIDE model.¹⁹ And documented patient-care hours are a requirement for health-professions admission,^{23,24} which is the incentive our workforce pathway uses; whether it converts into durable capacity is what Aim 2 tests.

Hurdles to adoption. The primary hurdle is local density at a cost the market sustains: families, providers, and caregivers must concentrate in the same market, and that market's revenue must cover the cost of bringing them together. Four specific hurdles follow. Families must trust an AI system with care decisions, which we address by drawing every answer from an expert-curated database audited by a blinded panel of licensed social workers in Aim 1, and by validating acceptance with family caregivers under IRB approval in Aim 2. Providers are wary of platforms, so they begin at no cost and are asked to pay only after value has been delivered, which is exactly how Aim 2 runs, with conversion and churn measured under real billing only afterward in Aim 3. New caregivers must be safe and welcome, which is why licensed providers employ, train, insure, and supervise them, and why Aim 2 measures the pathway end to end with both providers and workers. And every market must repeat affordably, which self-service portals and a documented market-entry playbook address, with acquisition cost per market measured in Aim 2 and re-measured in each Aim 3 wave.

INNOVATION

Key Innovation 1: capacity created, not moved. Every staffing channel serving this industry, whether an agency, a job board, or a gig platform, competes for workers already inside the direct-care labor market. One provider's hire is another provider's vacancy, and the national shortage does not shrink by a single worker. Olera adds people who were not in the field (Figure 3).

The first population tested is health-professions students, chosen because admission to their programs requires documented patient-care hours, which makes recruitment structurally inexpensive rather than dependent on wage competition. Health professions confer 263,800 undergraduate degrees a year,²³ and entry into nursing, medicine, and physician assistant training turns on documented direct patient care.²⁴ Our own pilot drew more than 900 applicants. New cohorts arrive every semester, so the pipeline refills rather than depleting.

The innovation is not the student channel, which any competitor could copy. It is the infrastructure underneath it: the machinery that converts a person who has never worked in eldercare into someone a licensed provider will hire, and that carries their verified record of hours, competencies, populations served, supervisor evaluations, and credential status forward across employers. That layer is what makes any new worker pool viable, and its value compounds with every shift worked. Task 2.5 tests a second, non-student pool against the same pathway in two campus-poor markets, converting extensibility from an assertion into a demonstrated result.

Key Innovation 2: execution, not information. The prevailing paradigm is that tools inform and families execute. A family is told which programs it may qualify for and is then left to prepare applications, gather

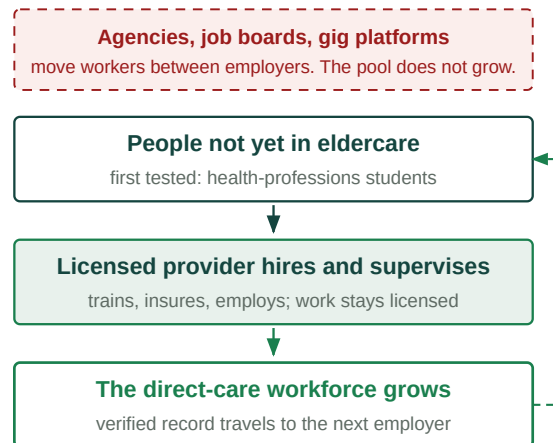


Figure 3. Existing channels move workers between employers. This pathway adds them, and the verified record follows the worker.

documents, file, wait, respond to requests, and confirm that care began. Those are precisely the steps at which families are lost, and no existing tool is accountable for them.

CareNavigator's agents perform the work and the family decides. **The unit of success changes from information delivered to care established.** That shift is what makes the outcome measurable, what makes it saleable to an organization that bears the cost of unmet need, and what makes the system improve: every completed case becomes supervised training data for the next one. The technical foundation, a multi-agent architecture over a retrieval layer with expert feedback, was built and evaluated under Phase IIB; what this award adds is execution and confirmation.

Key Innovation 3: a county-level record of what actually happens. Eldercare information exists in three layers. The open web holds public directories and program rules, often neither accurate nor current, and it is equally available to any general-purpose AI system. Beneath that sit local program realities and provider truths that were never published, which we assembled through field work, expert curation, and direct outreach into more than 72,000 records covering all fifty states. Beneath that sits a third layer that can only be learned by operating: what programs actually decide, county by county, where applications stall, whether capacity existed when it was needed, and what care ultimately started.

No public dataset holds the third layer. NHATS is a survey. Claims data record what was billed, not what was needed and refused. **What this project builds is the first longitudinal, county-level administrative record of what older adults need, what they qualify for, where the system fails them, and whether care was established.** Its commercial value is that a competitor who copies every feature begins with an empty database. Its scientific value is that this measurement does not currently exist for researchers or for the state agencies that plan around it, and we commit to publishing from it.

APPROACH

Overall research design. Olera builds two products, CareNavigator and Caregiver Staffing. During this award only Caregiver Staffing is sold and priced. Two NIH awards and our own capital have retired five commercial risks, which the CRP Progress Report documents. One remains, and research funding alone cannot retire it: **can a single local market pay for itself, and can we repeat it?**

The award answers that in three stages (Figure 4). **Aim 1** builds and verifies the technology both products require, in Year 1. **Aim 2** runs both products in two markets at no charge, with real families, real caregivers, and real providers, and measures whether care gets established and whether the staffing pathway completes. **Aim 3** opens six new markets, charges for staffing, and measures whether providers pay, whether the unit economics hold after real costs, and whether the playbook repeats.

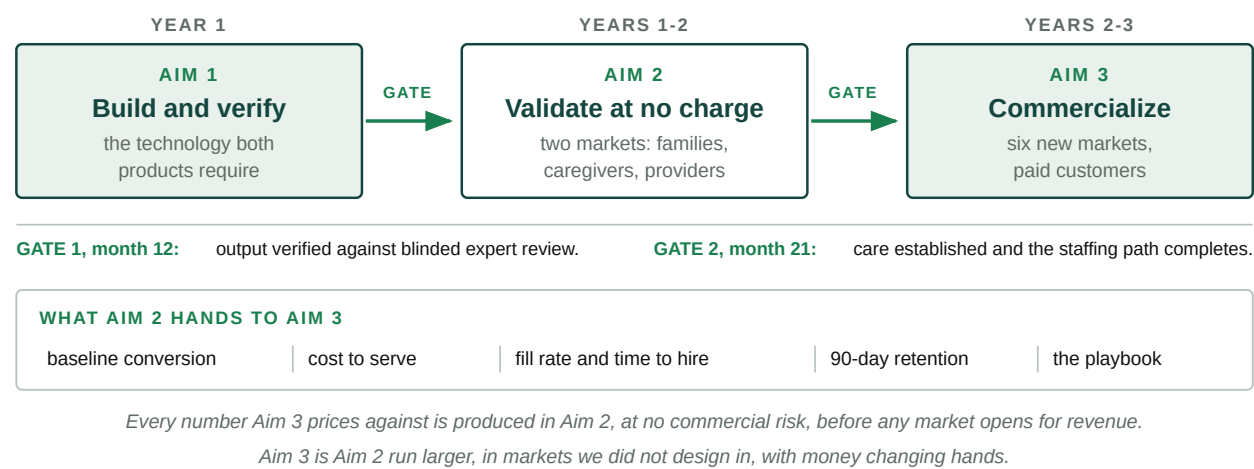


Figure 4. The three aims as one sequence. Each stage carries a gate, and Aim 2 produces every parameter that Aim 3 prices against. The sequence is the design, not a schedule. **Every parameter Aim 3 prices against is produced by Aim 2 at no commercial risk:** baseline conversion, cost to serve, fill rate and time to hire, 90-day worker retention, and the market-entry playbook itself. Aim 3 is Aim 2 run larger, in markets we did not design in, with money

changing hands. Each stage carries a gate, so failure is contained rather than propagated: if Aim 1 misses verification, Aim 2 does not activate; if Aim 2 misses establishment, wave one is held.

Primary endpoints throughout are operational or economic and come from platform, billing, and employment records. Human subjects research sits inside the aims as Tasks 2.3, 2.5, and 3.4, under Clemson University IRB approval with Dr. Fan as protocol lead. The project remains outside the NIH definition of a clinical trial by design: no participant is prospectively assigned to receive or not receive navigation, health-related outcomes appear only as secondary observational measures on a cohort that all receives the service, and the price conditions in Aim 3 are assigned to markets with purchase as the outcome. Formal determinations will be obtained rather than assumed, including for streams we expect to be non-human-subjects research.

Market architecture. The design sets the number of markets, not the reverse (Table 4). **Two markets in Aim 2**, run at no charge. One market cannot distinguish a product that works from a market that happens to work, so two are needed for any replication signal at all. They are chosen to differ on workforce source, one campus-rich and one campus-poor, because the riskiest assumption in this application is whether people new to care work can be recruited and retained. If that pipeline depends on proximity to a health-professions campus, we need to know while it is free to learn.

Six markets in Aim 3, all new, all paid. Price is assigned at the matched-market level rather than the account level, because providers in neighboring markets compare quotes, so the market is the unit of assignment and market count sets the precision of the price contrast. Three price points, two markets per arm, and within each arm one campus-rich and one campus-poor market, so price condition is crossed with market type rather than confounded with it. The six open in two waves of three: wave one at month 21, the only wave with runway to observe 12-month retention, and wave two at month 30, run from the written playbook by staff who did not design it. Socioeconomic status and urbanicity are measured and reported as stratification variables rather than made design axes. At roughly \$30,000 to enter a market, market entry is about six percent of the requested budget; the binding constraint is operating attention, not capital.

Requirement	What it demands of the design	Markets implied
Validate before spending	Real families, providers, and caregivers concentrated in one place, at no charge, so adoption is not confounded with price	2, run free in Aim 2
Separate the product from the market	One market cannot distinguish a product that works from a market that happens to work	2 rather than 1
Test the riskiest assumption while it is cheap	Whether new entrants can be recruited without a health-professions campus nearby	The 2 differ: campus-rich and campus-poor
Choose a price rather than guess one	Assignment at matched-market level because neighbors compare quotes; three price points to reveal shape, two markets per arm so an arm is not one market	6, paid, in Aim 3
Keep price unconfounded with market type	Within each arm, one campus-rich and one campus-poor market	3 arms crossed with 2 types
Show the playbook repeats	Two waves, the second run as written by staff who did not design it; wave one early enough to observe 12-month retention	Waves of 3 at months 21 and 30
Derived design	Two free validation markets, then six paid markets in two waves	8 markets, about 6% of the budget

Table 4. How the market design and its size are derived from what the aims require.

Specific Aim 1: Build and verify the technology both products require. (Year 1)

Central question: can we engineer the capabilities that make care establishment and caregiver placement work reliably enough to test commercially? Phase IIB built identification. This aim builds execution, the workforce infrastructure, and the data layer underneath both, and proves the output is good enough to put in front of families before any market is activated.

Task 1.1: Execution and follow-up, across aid and services. Families are lost at the same two points whether they are securing a benefit or securing care: while completing what the program or provider requires, and again while waiting for an answer. The loops built here handle both. For aid, that is assembling and submitting applications, tracking the documentation each program requires, and following the case to a decision. For services it is the same abstraction applied to a different counterparty: preparing and sending the

intake packet, scheduling the assessment, following up with more than one provider so a single decline does not end the case, and confirming the date care began.

Task 1.2: The aid, provider, and outcomes database, and the domain model built on it. Expansion and re-verification of the curated record, the instrumentation that captures what each completed case decided, and the domain-specific model trained on that record. *(Detailed scope owned by the PI; see Innovation.)*

Task 1.3: Caregiver workforce infrastructure and the verified experience record. Recruitment and screening workflows, training completion tracking, the provider-facing placement interface, and the verified record that accumulates hours, competencies, populations served, and supervisor evaluations and travels with the worker.

Task 1.4: Verification against blinded expert review. A panel of licensed clinical social workers, independent of the engineering team and holding no equity in Olera, audits a stratified random sample of cases each quarter. For each audited household the panel determines eligibility blinded to what the platform produced, and for a subset prepares the applications by hand. Both are compared with the platform's output, category by category for eligibility and field by field for applications. Agreement is reported as percent agreement with a 95 percent confidence interval and as Cohen's kappa, inter-rater reliability among panelists is established before any system output is adjudicated, and disagreements are routed to error analysis. Audited households are held out of tuning data. This is internal product verification, not human subjects research, and a determination will be obtained to that effect.

Metrics for Success and Quantitative Criteria (Aim 1):

Measure	Criterion	Source of the measurement
Agreement with the blinded expert panel	≥ 85%	Quarterly stratified audit, Task 1.4
Material errors against the expert gold standard	≤ 10%	Same audit
Inter-rater reliability among panelists	$\kappa \geq 0.70$	Established before adjudication
Execution loops complete end to end, aid and services	Both paths	Internal release testing
Scheduled follow-ups sent on their due date	≥ 95%	Platform telemetry
Verified experience record populated for every placement	100%	Workforce system

Table 5. Aim 1 quantitative success criteria.

Gate: the first two criteria must be met at month 12 before Aim 2 activates.

Potential problems and alternative strategies. Panel disagreement could reflect reviewer variability rather than system error, which is why inter-rater reliability is established first and reported alongside every accuracy figure. Program rules change during a three-year award and would degrade accuracy through no fault of the system; the maintenance pipeline re-verifies records on a rolling schedule and accuracy is reported separately for rules that changed inside the measurement window. If agreement stalls below the gate, the alternative is to narrow scope rather than widen it: restrict automated execution to the program and service categories where accuracy is highest, route the remainder to human-assisted execution, and report the boundary rather than an average that conceals it. Aim 2 activation is held until the gate is met, which is what the month 12 decision point exists to enforce.

Specific Aim 2: Validate both products in two markets, at no charge. (Years 1 to 2)

Central question: with real households, real providers, and real caregivers concentrated in one place, does care get established, does the staffing pathway complete end to end, and do all three sides accept it? This aim is where the commercial risk is removed. Nothing is charged, so adoption is not confounded with price, and every parameter Aim 3 needs in order to price is measured here first.

Task 2.1: Activate two markets. Two markets are opened, one campus-rich and one campus-poor, selected on existing organic family traffic, provider density, and state aid generosity. Families arrive through organic search, local partners including Area Agencies on Aging and Alzheimer's Association chapters, and direct outreach. Providers are recruited from the onboarded base and by direct outreach and are offered both products at no cost for the duration of the aim. Every activation is tagged by source, market, and cohort.

Task 2.2: Measure care establishment. The operational stream. Each household is followed from screening through the identified aid and services to a confirmed start date, with the step at which any household stops recorded. Reported outcomes are the share reaching established aid or care, time from first contact to established care, aid dollars secured, drop-off by step, and eligibility accuracy against the accumulating volume of executed cases, which tests whether the database sharpens with use. This is platform telemetry, not human subjects research.

Task 2.3: Validate CareNavigator with family caregivers. (*Human subjects research; Clemson University IRB.*) Twenty-five family caregivers of people living with dementia complete a mixed-methods evaluation of the integrated product, following the User Experience Evaluation in Intelligent Environments framework, with the sample size informed by our prior work. In a 60-minute moderated session each participant completes a needs assessment, reviews eligibility matches, prepares one application package, and locates the status of a filed application, thinking aloud while the moderator records task success against pre-defined criteria, time on task, errors, and assists. Post-use measures are the System Usability Scale and the 12-item trust-in-automation scale, followed by a semi-structured interview on where the system lost them. Surveys at four weeks and three months record whether the participant reached aid or care. Transcripts are analyzed thematically against a shared codebook by two independent coders with inter-coder agreement reported, and findings are integrated in a joint display that converts into an engineering backlog.

Task 2.4: Recruit, place, and retain caregivers. Recruitment runs through campus advisors, health-professions organizations, and community channels. Prior direct-care employment is ascertained at intake, before placement, so the share of workers new to the field is measured rather than inferred. Each placement is timestamped from application through screening, provider interview, hire, and first confirmed shift, which yields time to hire and fill rate. Retention is reported by cohort at 90 days, and separately for workers who remain with the placing provider and those who move to another licensed employer carrying their verified record, because a worker who changes employers still represents capacity added to the field. Placed workers are also invited into the interview component of Task 2.5; research participation is firewalled from employment decisions, shift allocation, and any commercial conversation, and consent language states this.

Task 2.5: Validate the staffing product with providers and workers. (*Human subjects research; Clemson University IRB.*) A sequential mixed-methods implementation study in two phases. The formative phase enrolls approximately 30 provider participants across owner, recruiter, and scheduler roles and approximately 20 placed workers, who complete role-specific tasks against pre-defined criteria and then a guided interview on workflow fit, supervision and safety, training needs, and priorities for improvement. Recruitment continues in small batches until no new high-priority issue emerges in two consecutive batches. The field phase follows the refined product through live semester cohorts in both markets. Measures are the System Usability Scale and the Acceptability, Appropriateness, and Feasibility of Intervention Measures, reported with 95 percent confidence intervals as implementation estimates rather than between-group tests. A stated-preference component, administered apart from any sales conversation, seeds the price range Aim 3 tests.

Task 2.6: Measure cost to acquire and cost to serve. Acquisition cost is measured per family and per provider by channel and market. Cost to serve is measured per established case and per placed worker using time-driven activity-based costing over the compute behind the agents, the messages the loops send, navigator time, and support time. Both formulas are fixed before measurement begins and both read from live records. These figures are the denominators of every economic claim in Aim 3.

Metrics for Success and Quantitative Criteria (Aim 2):

Measure	Criterion	Source of the measurement
Households reaching established aid or care	Reported with 95% CI	Confirmed start date, Task 2.2
Outcome ascertainment at the pre-specified window	≥ 80%	Task 2.2
Usability, System Usability Scale	≥ 72	Task 2.3, IRB, n = 25
Trust in automation	≥ 5 of 7	Task 2.3, IRB
Placed workers with no prior direct-care employment	Reported	Ascertained at intake, Task 2.4
Worker retention at 90 days	Reported by cohort	Employment and verified-record data
Provider acceptability, appropriateness, feasibility	≥ 4.0 of 5	Task 2.5, IRB

Measure	Criterion	Source of the measurement
Cost to acquire and cost to serve	Measured, both sides	Task 2.6, time-driven activity-based costing

Table 6. Aim 2 quantitative success criteria.

Gate: care established and the staffing path completing end to end, at month 21, before wave one opens.

Potential problems and alternative strategies. Worker retention is the most consequential risk in this application. If people new to the field leave within 90 days, providers will not pay a second time and the commercial model in Aim 3 weakens regardless of how well navigation performs. Two responses are pre-specified. Retention is reported both with the placing provider and in direct care with any licensed employer, because those answer different questions and only the second speaks to capacity added. If retention with the placing provider is poor while workforce retention holds, the product emphasis shifts from placement to the verified record itself, which providers value as reduced screening burden. Campus seasonality could create hiring gaps between terms, so cohorts are staggered across the two markets. If the campus-poor market produces no caregivers, that is a finding rather than a loss: the family side continues there and Aim 3's market selection is revised toward the conditions that worked, with the boundary stated. If family volume in either market is too low for providers to experience value, family acquisition is reallocated before any provider-side conclusion is drawn. The pathway is population-agnostic by design, and a non-student worker pool is the first extension after the award.

Specific Aim 3: Determine whether providers pay and the market model repeats. (Years 2 to 3)

Central question: will providers pay for staffing, do the unit economics hold after real costs, and does the playbook repeat in markets we did not design in? Aims 1 and 2 establish that the products work. This aim tests whether that value converts into recurring revenue at a price and a cost structure that sustain a market, which is the condition on which the business, and any private investment in it, depends.

Task 3.1: Open six markets in two waves. Wave one opens three markets at month 21 and wave two three more at month 30. Both run the playbook documented in Aim 2; wave two is executed by staff who did not write it, with deviations logged. Waves are compared on time to first provider, cost per activation, and the share of activations that are self-service rather than staffed, which tests whether the model repeats affordably rather than merely repeats.

Task 3.2: Set price and packaging under real billing. Three price points are tested, seeded by the stated-preference work in Task 2.5 and anchored on the prices providers have already paid us, approximately \$275 per month and approximately \$150 per placement. Conditions are assigned at the matched-market level rather than the account level, because providers in neighboring markets compare quotes; markets are paired non-adjacently and a contamination monitor runs in the offer workflow. Price points, package variants, the primary contrast, the follow-up horizon, and the decision rule are pre-registered before any outcome is reviewed. The primary outcome is paid conversion within 60 days of offer, analyzed with generalized estimating equations at the account level with market as the cluster and small-sample corrections. With six markets, arm-level contrasts are estimated with confidence intervals rather than significance-tested, which is the honest use of this design. A pre-registered interim analysis at month 30 drops a dominated arm and reallocates to wave two. The decision rule selects the price maximizing expected 12-month revenue per account rather than conversion alone, because a price that converts and then churns is worse than one that does neither.

Task 3.3: Measure unit economics and retention. Every figure comes from live billing, payroll, and cost records. Customer acquisition cost includes spending on non-converters. Monthly net account margin is revenue less the cost to serve defined in Task 2.6. Retention is estimated at 3, 6, and 12 months. Time to churn is analyzed with discrete-time survival models on account-month records matching the monthly billing cycle, with voluntary cancellation, uncured payment failure, and downgrade treated as competing events, and cause-specific hazards and cumulative incidence reported. Restricted mean survival time at 12 months gives expected account lifetime, and lifetime multiplied by monthly net margin gives lifetime value with a confidence interval, so the payback and lifetime-to-acquisition-cost criteria carry measured uncertainty rather than point estimates. Per-market contribution margin is reported after the cost of serving that market's families, which is the number that determines whether a market pays for itself.

Task 3.4: The Aim 2 instruments at scale. *(Human subjects research; Clemson University IRB.)* The implementation measures from Task 2.5 are administered across the paid cohort, with a baseline at enrollment

and follow-up at four weeks and three months or after a key event such as activation, downgrade, or churn, with approximately 150 provider decision-makers expected. Semi-structured interviews with converters, non-converters, and churned accounts examine what was purchased, what value was perceived, how price was judged, and why accounts lapsed. Baseline perceived value enters the Task 3.3 survival models as a pre-specified predictor of churn. The person-level participant and the account-level economic unit are defined separately throughout.

Task 3.5: Independent rebuild and the investor evidence package. An independent analyst engaged under subcontract, working from the pre-specified analysis plan and the raw billing, payroll, and cost records, rebuilds revenue, acquisition cost, cost to serve, margin, retention, and per-market profitability without access to our operating model. The analyst does not set price or packaging. Discrepancies between the rebuild and the operating model are investigated and reported rather than silently reconciled. The package assembles the confirmed model into two forward trajectories, expansion funded by reinvested margin and accelerated expansion under private capital, and is the evidence base for the Commercialization Plan's fundraising strategy.

Metrics for Success and Quantitative Criteria (Aim 3):

Measure	Criterion	Source of the measurement
Paid conversion within 60 days, by price arm	Estimated with 95% CI	Task 3.2, matched-market assignment
Lifetime value against acquisition cost, 12 months	$\geq 3 : 1$	Task 3.3, restricted mean survival time
Payback period on acquisition cost	< 12 months	Task 3.3
Per-market contribution margin after cost of serving families	Positive	Task 3.3
Retention at 3, 6, and 12 months	Reported by wave	Discrete-time survival, competing events
Cost per activation, wave two against wave one	Falling	Task 3.1
Independent rebuild delivered, discrepancies reported	Delivered	Task 3.5

Table 7. Aim 3 quantitative success criteria.

Potential problems, alternative strategies, and the stop rule. Contamination across neighboring markets is the standard objection to market-level assignment; the pre-registered mitigations are non-adjacent pairing, a contamination monitor in the offer workflow, and account-level assignment specified in advance as a sensitivity analysis. Conversion could prove too low at every price for the arms to be distinguishable, in which case the interim analysis reallocates and the finding is reported as a finding. If price sits below cost to serve, the pre-registered alternatives run in order: improve the product where the Task 3.4 interviews locate the gap, reduce acquisition and serving cost, repackage, and re-test in wave two. If margins are positive but below the level a market needs to fund itself, no further markets open until the cost structure improves. If wave two slips, wave one still carries the pricing, unit economics, and 12-month retention results, and replication is reported on the markets that opened.

The stop rule. If, at the month 30 interim analysis, fewer than 40 percent of placed workers remain in direct care at 90 days *and* paid conversion is below 20 percent at every offered price, we will report the provider-funded model as disconfirmed, hold wave two, and redirect the remaining effort to completing and publishing the analysis. A commercialization program should fund projects willing to find out they are wrong.

If staffing revenue alone proves insufficient. The reviewer's fair question is whether one product can carry a market. Two-thirds of home-care providers turned away business in 2023 for lack of staff,³ so the demand is not in doubt, and the price sits far below what those providers already pay to hire. Should the measured economics still fall short, Olera holds assets this award produces that support adjacent revenue without new invention: the provider growth tools already operating today, an employer-paid caregiver benefit on the same navigation product, contracted navigation for risk-bearing organizations on the precedent of the Medicare GUIDE model,¹⁹ and aggregate market intelligence from the outcomes record. **None is tested in this award and none carries an endpoint here.** They are modeled in the Commercialization Plan.

Technical assistance and project oversight. Three external providers carry defined deliverables. **Clemson University**, through Dr. Fan as Co-Investigator at 25 percent effort, designs and executes the human subjects studies in Tasks 2.3, 2.5, and 3.4 and holds the protocols, integrated through joint quarterly protocol review and

shared data-management procedures specified in the subaward. **An independent financial analyst** performs Task 3.5, and **pricing and actuarial consultation** supports the pre-registration in Task 3.2. The Principal Investigator holds integration authority and all gate decisions, supported by a weekly cross-aim operating review and a quarterly milestone review against Tables 5 through 7. Consultant scopes and reporting cadence are specified in the Project Management Plan.

Regulatory plan. No federal premarket pathway applies. CareNavigator is navigational and administrative software that does not diagnose, treat, or make clinical decisions, and is therefore not a medical device; no premarket authorization gates commercialization and nothing in this plan waits on a federal approval. Four regimes govern and are addressed operationally: family data under HIPAA-aligned safeguards; referral practice, where charging no fee means no steering incentive exists and federally reimbursed providers can participate; employment, training, insurance, and supervision of every placed worker, which rest with the licensed provider rather than with Olera; and any joining of an external record to ours, which requires an executed data-use agreement and an IRB determination.

Timetable and what the award leaves behind. Figure 5 gives the quarter-by-quarter schedule and the four decision points. At month 12 the Aim 1 gate must be met or market activation is held. At month 21 the Aim 2 gate must be met or wave one is held. At month 30 the interim price analysis runs and the stop rule is evaluated before wave two opens. At award end the independent rebuild is delivered.

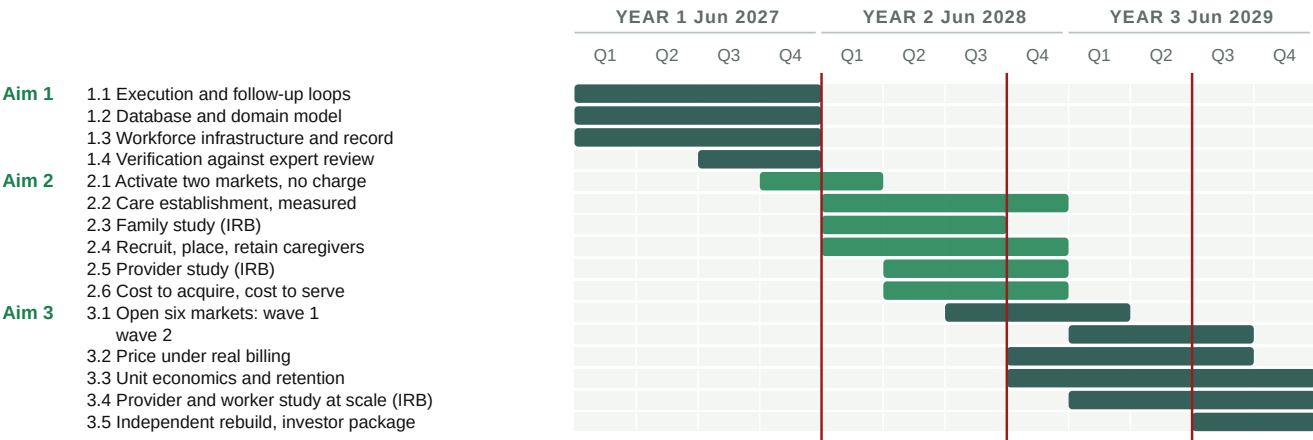


Figure 5. Three-year timetable, showing the sequencing among the aims, staged market entry, and the four formal decision points.

Successful completion leaves Olera holding four things that do not exist today: a product verified against blinded expert review, a price chosen by experiment rather than assumption, unit economics an outside analyst rebuilt from records, and a market-entry playbook run as written in six markets by people who did not write it. That is the evidence a private investor requires to fund national expansion, and it is the operational record on which the adjacent revenue lines in the Commercialization Plan rest.

CRP PROGRESS REPORT

Development status of the technology. CareNavigator operates in production today. Families screen their needs and means, receive matched aid programs and providers drawn from an expert-curated national database of more than 72,000 records covering all fifty states, and receive AI-drafted, expert-approved guidance. The multi-agent navigation layer, developed and evaluated under Phase IIB using retrieval-augmented generation, parameter-efficient fine-tuning, and reinforcement learning from expert feedback, is in production integration during the remaining Phase IIB year. Execution and follow-up, the capability Aim 1 builds, do not exist in the product today and are not funded by any existing award.

What Phase I through IIB established. The technology was developed across NIA SBIR Phase I/II Fast-Track and Phase IIB awards (1R44AG074116), which received impact scores of 20 and 25 respectively. Four peer-reviewed studies evaluated the platform with family caregivers: usability scored 4.57 of 5, and technology acceptance scored 5.83 of 7 after four weeks of independent use among 65 caregivers, with higher use frequency associated with higher acceptance.^{25,26,27,28} The multi-agent version scored 5.73 of 7 among 31 dementia caregivers in a study now in preparation.²⁹ The Phase IIB review assessed the platform's commercial potential as extremely high, citing its ability to help caregivers access eligible funding.

Commercialization progress, including work beyond the Phase II scope. Table 8 records what each activity retired and which funding produced it. Two of the five risks were retired outside the Phase II scope, using I-Corps support and the company's own capital, which is the record of commercialization initiative this program is designed to reward.

Risk	Retired by	Evidence
Technical Could the data be assembled?	NIA Phase I to IIB	An expert-curated national database of more than 72,000 aid programs and providers across all fifty states, and a multi-agent eldercare AI now entering production integration.
User Would families accept it?	NIA Phase I to IIB	Four peer-reviewed evaluations with family caregivers: usability 4.57 of 5; acceptance 5.83 of 7 after four weeks of use (n = 65); multi-agent version 5.73 of 7 (n = 31, in preparation).
Demand Would families come, and at what cost?	Beyond Phase II scope; company funds	Organic traffic grew from roughly 50 visits a day in 2023 to more than 500 today, about 15,500 a month, from nearly every county in the country at near-zero acquisition cost.
Supply Would providers participate?	I-Corps and company funds	More than 300 customer-discovery conversations with owners and operators through NIH and NSF I-Corps. More than 700 providers have since claimed a listing at no charge, roughly 150 more each month.
Workforce and willingness to pay Could we recruit caregivers, and would anyone pay?	Beyond Phase II scope; company funds	A student caregiver pilot drew more than 900 applicants and placed more than 20 into provider jobs. Four providers trialed the staffing product and three paid, at roughly \$50 to \$275 per placement and \$275 per month, across multiple semesters. Delivery was manual, which capped scale; Aim 1 removes the cap.
The risk that remains Can one local market pay for itself, and can we repeat it?	Not yet retired	Measurable only with real customers paying real prices over long enough to observe churn. This is the uncertainty the three aims remove.

Table 8. Commercialization risks retired to date, the funding that retired each, and the risk that remains.

Ongoing Phase IIB work this award does not re-fund. The remaining Phase IIB year completes production integration of the agent system and continues database expansion. This application begins where that work ends and duplicates none of it. No budget line here supports agent development already funded under Phase IIB; Aim 1 builds execution, follow-up, confirmation, and the workforce infrastructure, none of which is in scope for the existing award.

The team that will execute this work. Dr. Falohun has served as Principal Investigator on both prior NIA awards, which were scored at 20 and 25 and delivered against their milestones. Dr. DuBose, a physician with an MBA, leads product and market operations and built the organic acquisition channel and the provider pipeline described above. Dr. Fan, at Clemson University, serves as Co-Investigator at 25 percent effort and leads all human subjects protocols; she is a co-author on the platform evaluation record cited here. Dr. Marcia Ory of Texas A&M University, whose NIH career spans aging services and implementation research, advises on study design and dissemination and is likewise a co-author on that record. David Qu advises on commercialization and investor readiness and has pressure tested the commercial plan in this application.

Olera operates as a fully distributed company: two full-time and two part-time employees, dedicated engineering and marketing support, a two-person care navigation call center, and contract staff, scaling with this award into market and workforce operations roles. Being distributed is why an eight-market design is feasible at this size: no market requires a local office.

The risk that remains. Nine years of federal and company investment retired every commercial risk that could be retired without customers. What none of it purchased is proof that a local market pays for itself and that the result repeats, because that cannot be purchased. It has to be measured in real markets, with real money changing hands, over long enough to observe churn. **That is the uncertainty the three aims remove, and it is the reason this application exists.**

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